

Module End Assignment:

Data Transformation, Modeling & Visualization with Power BI

1. DATA SOURCE

- File: bike-stations-sharing-data
- Total Records: 2856
- Total Columns: 12
- Columns: Number, Name, Address, Position, Banking, Bonus, Status, Contract Name, Bike Stands, Available Bike Stands, Available Bikes, Last Update

2. DATA CLEANING

- Removed extra spaces using TRIM and CLEAN function in power query editor
- Ensure the font size using UPPER CASE

3. DATA TRANSFORMATION

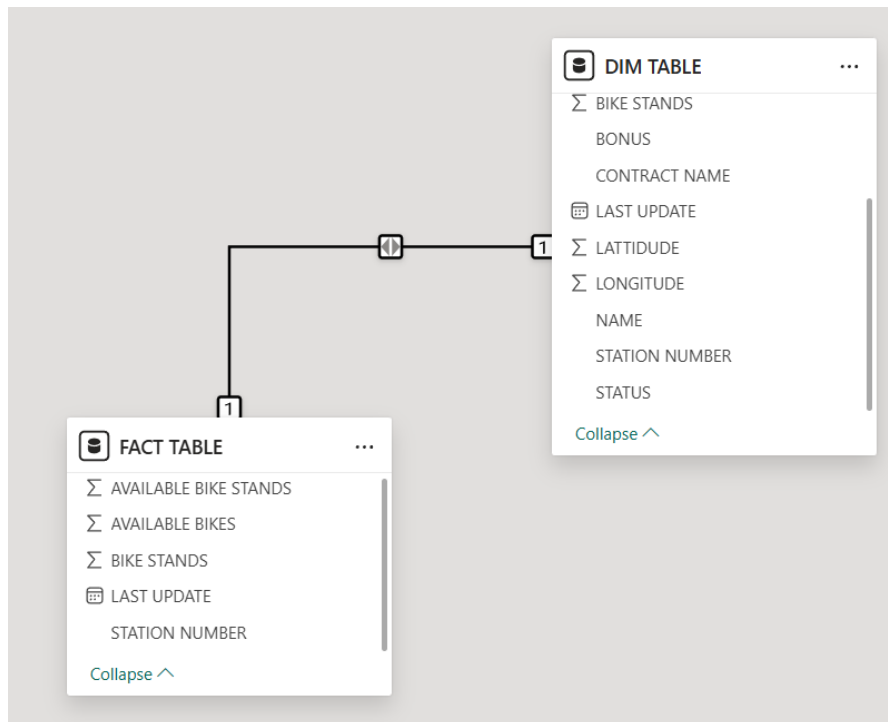
- Converted numeric fields to Whole Number and text fields to Text.
- Converted Last Update column to Date/Time and extracted Date for analysis.

4. DUPLICATE REMOVAL

- Removed duplicate records based on STATION NUMBER using Power Query:
Home → Remove Rows → Remove Duplicates.

5. DATA MODEL

- Fact Table: Station Number, Bike Stands, Available Bike Stands, Available Bikes.
- Dimension Table: Station Number, Station Name, Address, Contract Name, Status, Bike Stands, Available Bike Stands, Available Bikes.



6. DESCRIPTIVE ANALYSIS

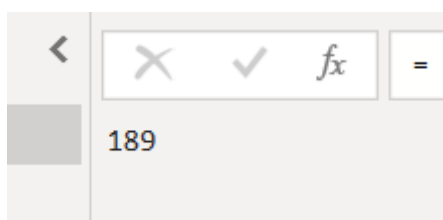
- **Total Bike Stations:** 1002
- **Total Bike Stands:** 18792
- **Available Bikes:** 5777
- **Available Empty Stands:** 10048
- **Average Bikes per Station:** 5.77

TOTAL BIKE STANDS	TOTAL STATION COUNT	TOTAL AVAILABLE STANDS	TOTAL AVAILABLE BIKES	AVG BIKE PER STATION
18792	1002	10048	5777	5.77

7. DIAGNOSTIC ANALYSIS

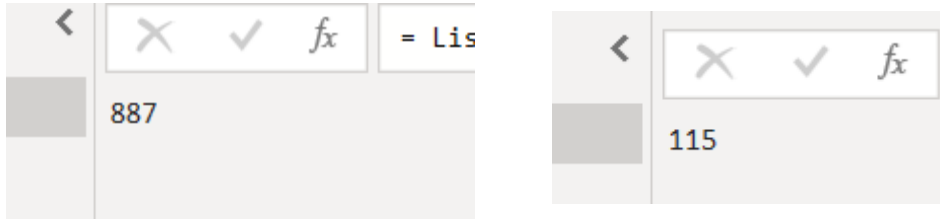
Banking enables options:

- Banking option Yes-189, No - 813



Station status:

- Total station Opened -887, Closed -115



The image shows two calculator interface snippets. The left snippet displays the number 887 in red, representing the total stations opened. The right snippet displays the number 115 in red, representing the total stations closed. Both calculators have a top bar with a back arrow, a clear button (X), a checkmark button (✓), a function button (fx), and an equals button (=). The main display area shows the respective numbers.

8. PREDICTIVE ANALYSIS

Based on patterns:

- High demand stations may **run out of bikes frequently**
- Some stations may **remain underutilized**
- Peak demand likely during:
 - Morning commute
 - Evening return hours

If trends continue:

- Bike shortages will increase in busy locations.

9. PRESCRIPTIVE ANALYSIS

- **Bike Rebalancing**
Move bikes from **low-demand stations** → **high-demand stations**.
- **Add More Bike Stands**
Stations with frequent shortages should have:
additional docks
higher bike capacity.
- **Predict Demand**
- Use **AI / demand forecasting models** to predict:
busy hours
bike shortages.
- **Maintenance Monitoring**
Track stations that frequently become **closed**.

10.DASHBOARDS

